THE TWENTIETH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

November 21, 1959

Morning Session

1. Let n be a positive integer. Prove that $x^n - (1/x^n)$ is expressible as a polynomial in $x - (1/x)$ with real coefficients if and only if n is odd.
(page 497)

2. Prove that if the points in the complex plane corresponding to two distinct complex numbers z_1 and z_2 are two vertices of an equilateral triangle, then the third vertex corresponds to $-\omega z_1 - \omega^2 z_2$, where ω is an imaginary cube root of unity. (page 498)

3. Find all complex-valued functions f of a complex variable such that $f(z) + zf(1 - z) = 1 + z$ for all z. (page 499)

4. If f and g are real-valued functions of one real variable, show that there exist numbers x and y such that $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $|xy - f(x) - g(y)| \geq 1/4$. (page 499)

5. A sparrow, flying horizontally in a straight line, is 50 feet directly below an eagle and 100 feet directly above a hawk. Both hawk and eagle fly directly toward the sparrow, reaching it simultaneously. The hawk flies twice as fast as the sparrow. How far does each bird fly? At what rate does the eagle fly?
(page 500)

6. Let m and n be integers greater than 1, and $a_1, \ldots, a_{m+1}$ real numbers. Prove that there exist real n by n matrices $A_1, \ldots, A_m$ such that (i) $\operatorname{Det}(A_j) = a_j$ for $j = 1, \ldots, m$, and (ii) $\operatorname{Det}(A_1 + \cdots + A_m) = a_{m+1}$.
(page 503)

7. If f is a real-valued function of one real variable which has a continuous derivative on the closed interval $[a, b]$ and for which there is no $x \in [a, b]$ such that $f(x) = f'(x) = 0$, then prove that there is a function g with continuous first derivative on $[a, b]$ such that $fg' - f'g$ is positive on $[a, b]$.
(page 504)

Afternoon Session

1. Let each of m distinct points on the positive part of the X-axis be joined to n distinct points on the positive part of the Y-axis. Obtain a formula for

the number of intersection points of these segments (exclusive of endpoints), assuming that no three of the segments are concurrent. (page 505)

2. Let c be a positive real number. Prove that c can be expressed in infinitely many ways as a sum of infinitely many distinct terms selected from the sequence

$$1/10,\ 1/20,\ \ldots,\ 1/10n,\ \ldots. \qquad \text{(page 506)}$$

3. Give an example of a continuous real-valued function f from $[0, 1]$ to $[0, 1]$ which takes on every value in $[0, 1]$ an infinite number of times. (page 507)

4. Given the following matrix of 25 elements

$$\begin{bmatrix} 11 & 17 & 25 & 19 & 16 \\ 24 & 10 & 13 & 15 & 3 \\ 12 & 5 & 14 & 2 & 18 \\ 23 & 4 & 1 & 8 & 22 \\ 6 & 20 & 7 & 21 & 9 \end{bmatrix},$$

choose five of these elements, no two coming from the same row or column, in such a way that the minimum of these five elements is as large as possible. Prove that your answer is correct. (page 510)

5. Find the equation of the smallest sphere which is tangent to both of the lines: (i) $x = t + 1, y = 2t + 4, z = -3t + 5$, and (ii) $x = 4t - 12, y = -t + 8, z = t + 17$. (page 511)

6. Prove that, if x and y are positive irrationals such that $1/x + 1/y = 1$, then the sequences $[x], [2x], \ldots, [nx], \ldots$ and $[y], [2y], \ldots, [ny], \ldots$ together include every positive integer exactly once. (The notation $[x]$ means the largest integer not exceeding x.) (page 513)

7. For each positive integer n, let f_n be a real-valued symmetric function of n real variables. Suppose that for all n and for all real numbers $x_1, \ldots, x_{n+1}, y$, it is true that

(1) $f_n(x_1 + y, \ldots, x_n + y) = f_n(x_1, \ldots, x_n) + y$,

(2) $f_n(-x_1, \ldots, -x_n) = -f_n(x_1, \ldots x_n)$,

(3) $f_{n+1}(f_n(x_1, \ldots, x_n), \ldots, f_n(x_1, \ldots, x_n), x_{n+1}) = f_{n+1}(x_1, \ldots, x_{n+1})$.

Prove that

(4) $f_n(x_1, \ldots, x_n) = (x_1 + \cdots + x_n)/n$. (page 515)